
Noushad Ali Malawat

Pondicherry University

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About Me

Physics graduate from Pondicherry University with hands-on research experience in spectroscopy, materials characterization, and experimental data analysis. At BARC Mumbai, I worked on high-pressure Raman spectroscopy using diamond anvil cells, gaining experience in optical instrumentation, pressure calibration and spectral analysis across multiple material systems. At IIT Tirupati, I synthesized and characterized nickel–cobalt hydroxide electrodes using XRD and electrochemical techniques. I also worked on stellar spectroscopy at the Indian Institute of Astrophysics, developing a data-reduction pipeline using PyRAF. My MSc dissertation focused on compound nucleus decay using the Dynamical Cluster-decay Model and I co-authored a paper presented at the DAE Symposium on Nuclear Physics. I am looking for a PhD position in experimental condensed matter physics or materials characterization where spectroscopy and structure–property relationships are central to the research.

Experience

1. Pondicherry University (MSc Dissertation) [{View Report}](#) 01 Nov. 2025 – 30 April 2026
Kalapet, Puducherry, India
 - Decay of the Compound Nucleus $^{44}\text{Ti}^*$ Formed in the $^{32}\text{S} + ^{12}\text{C}$ Reaction at $E_{\text{CN}}^* \approx 71$ MeV
 2. Indian Institute of Astrophysics (IIA) [{View Report}](#) 22 Dec. 2025 – 09 Jan. 2026
Vainu Bappu Observatory (VBO), Kavalur, Tamil Nadu, India
 - Estimating stellar parameters of bright stars using low-resolution spectra
 3. Bhabha Atomic Research Center (BARC) [{View Report}](#) 08 May 2025 – 30 June 2025
HP & SRPD • Mumbai, Maharashtra, India
 - A vibrational analysis of compound SrCO_3 under high pressure
 4. Indian Institute of Technology (IIT) Tirupati [{View Report}](#) 15 May 2024 – 12 July 2024
Andhra Pradesh, India
 - Nickel–Cobalt Hydroxide based supercapacitor -: Synthesis, characterization and electrochemical evolution
 5. B.K. Birla college Kalyan March 2024
Mumbai University • Kalyan, Maharashtra, India
 - Alcohol based thermometer, which is used in polar regions
 6. B.K. Birla College Kalyan October 2023
Mumbai University • Kalyan, Maharashtra, India
 - Distance Measuring Device Using Lasers
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Workshop and Conferences

- Presented a paper titled “Fragmentation Profiles of Light and Heavy Compound Nuclei Formed in $^{12-15}\text{C}$ Induced Reactions ” at the National Conference on Science, Engineering and Technology for a Sustainable Future (SETSF-2026), held at Sri Guru Granth Sahib World University, Punjab, February 26–27, 2026.
- Participated as a delegate in the National Workshop on Nuclear–Hydrogen Synergy (NWNHS-2026), focusing on advanced nuclear energy systems and hydrogen-based sustainable technologies, organized by the Department of Green Energy Technology, Pondicherry University, Puducherry, India, January 30, 2026.
- Presented a paper at: The 69th DAE Symposium on Nuclear Physics, organized by the Board of Research in Nuclear Sciences (BRNS), Government of India, held at National Institute of Technology (NIT) Jalandhar, Punjab, India, December 08–12, 2025.
- 11th Aditya -L1 Support Cell Workshop (06-08 Oct 2025)
Organized by the Department of Physics, Pondicherry University, Puducherry, in association with Indian Space Research Organisation (ISRO), Bangalore & Aryabhata Research Institute of Observational Sciences (ARIES), Nainital
- National Workshop on Advances in Nuclear Energy for Sustainable Environment (NWANES-2025)
Pondicherry University, Puducherry, in Collaboration with the Department of Atomic Energy, Indira Gandhi Centre for Atomic Research (IGCAR), GOVERNMENT OF INDIA
- Summer School on Nuclear Physics (5-11 July 2025)
Ibrahim Mirza,
University of Tennessee, USA
- Summer Conference on High Energy Physics and Astrophysics Ninth Edition (10-11 May 2025)
Ibrahim Mirza,
University of Tennessee, USA
- Workshop on astrophysical dust ices insights from recent telescopes (10 March 2025)
Physical Research Laboratory, Ahmedabad, India

Publications

- Navjot Kaur, Noushad Ali Malawat, Rupinder Kaur, Sukhjeet Singh, and Birbikram Singh.
Comparative decay analysis of compound nuclei $^{24-27}\text{Mg}^*$ and $^{244-247}\text{Cm}^*$ formed in $^{12-15}\text{C}$ induced reactions.
Proceedings of the DAE Symposium on Nuclear Physics, Vol. 69, 2025. (*Presented*)

Activities

- PRAVAH 2022-2023 (Intra-college competition)
2nd position in the science exhibition.
 - National Science Day (Intra-college competition) Organized by Science Association 2022-2023
1st position in the project demonstration of laser show devices
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Presentation

- Nuclear Fusion -: An application of quantum tunneling (Summer school on nuclear physics 5-11 July 2025)

Education

MSc (Physics) {7.15/10 CGPA} 05/2026
Pondicherry University • Kalapet, Puducherry, India

BSc (Physics) {9.3/10 CGPA} 05/2024
B.K. Birla College Kalyan, Mumbai University • Kalyan, Maharashtra, India

HSC (12th) —RBSE {95.80 %} 05/2021
Marwar Montessori Senior Secondary School • Makrana, Rajasthan, India

SSC (10th) — RBSE {82.17 %} 05/2019
Marudhar Montessori Secondary School • Makrana, Rajasthan, India

Skills

- Python, Forton, Overleaf (LaTeX)
 - Raman Spectroscopy (Under high pressure), Electrochemical Workstation (performed Cyclic Voltammetry and Galvanostatic Charge – Discharge), Bentch Top X-Ray Diffraction, Telescope (Handled STAR TRACKER 150/750 EQ Reflector Telescope), and basic electronic equipment
 - Origin, PeakFit
 - Good Communication skills, Analytic Thinking, and Team leading
 - English, Hindi, Marwari, Urdu
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Extra-curricular activities

- National Science Day (Intra-college competition) Organized by Science Association 2022-2023, 1st prize in the scientific rangoli competition
 - Umesh Sir's Training center (Monthly competition), 1st prize in Skit - War
 - Organized National Science Day in the Physics Department, B.K. Birla College, Kalyan, Mumbai University
 - Active member of Pondicherry University and B.K. Birla Astronomy Club
 - Writing poetry in Urdu and managing a social media channel to share it
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